Cascade Pruning Networks: Learning When to Stop Re-ranking

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Abstract

Neural re-rankers dominate the cost of modern search. We present Cascade Pruning Networks, which learn calibrated exit decisions under a latency budget. CPN cuts p95 latency by 77% on MS MARCO with a negligible loss of 0.004 in nDCG@10, and reduces GPU cost by 63% in a production e-commerce deployment.

1 Introduction

Long-running retrieval systems waste most of their compute re-ranking documents that a cheap signal could have discarded. We study this waste in production search and observe that 71% of candidates sent to the neural re-ranker are never shown to a user.

We propose Cascade Pruning Networks (CPN), a family of models that learn when to stop. Each stage emits a calibrated confidence, and a candidate leaves the cascade as soon as its fate is certain.

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2 Related Work

Early-exit networks were introduced for image classification. Learning-to-rank cascades predate neural rankers but rely on hand-tuned thresholds. CPN differs in that thresholds are learned jointly with the rankers, under an explicit latency budget.

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3 Method

Let f1, f2 and f3 denote rankers of increasing cost. For candidate d and query q, stage k produces a score s and a confidence c. A candidate exits at stage k when c exceeds a learned threshold t_k.

Training minimizes a ranking loss plus lambda times the expected latency. The expected latency is differentiable because exit probabilities are relaxed with a temperature-scaled sigmoid during training and hardened at inference time.

We calibrate confidences with isotonic regression on a held-out split of 50,000 queries, which keeps the exit decision stable under distribution shift.

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4 Experiments

We evaluate on MS MARCO v2 and on a private e-commerce log of 12 million queries. Baselines are a single cross-encoder, a BM25 + cross-encoder pipeline, and a hand-tuned cascade.

On MS MARCO, CPN reduces p95 latency from 182 ms to 41 ms, a reduction of 77%, while nDCG@10 falls by only 0.004. On the e-commerce log, GPU cost falls by 63%, and an online A/B test

over 14 days showed no significant change in click-through rate.

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5 Limitations

CPN assumes that stage costs are stable, and our calibration degrades when the corpus changes quickly. In a simulated news corpus with daily churn, the exit error doubled after nine days unless the thresholds were retrained. We did not study adversarial queries.

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6 Conclusion

Most re-ranking compute is spent on decisions that are already certain. Learned early exit recovers that compute at almost no cost in quality. The main open problem is keeping calibration fresh without labelled data.

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